MANAS REDDY ARUMALLA

Robotics & Control Engineer | MSc Autonomy Technologies, FAU Erlangen-Nürnberg
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SUMMARY

Control & autonomy engineer specializing in estimation and safe control for aerial and mobile robots. Published author (ICRM 2025) and DST-funded researcher, with a research-grade portfolio spanning a vision-only drone that lands on moving ship decks, a bimanual manipulation platform, and 15+ benchmarked controller designs. Comfortable across the full stack — dynamic modelling and control design in MATLAB, validation in MuJoCo/Gazebo, and deployment to ROS 2 and embedded hardware. Interested in perception-driven, provably safe control for autonomous systems.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg *2026 – Present*
M.Sc. Autonomy Technologies — Nürnberg, Germany

Amrita Vishwa Vidyapeetham *2021 – 2025*
B.Tech, Automation & Robotics — Coimbatore, India | CGPA 8.84/10.0 (First Class with Distinction)
Relevant coursework: Control Systems, Robotics & Mobile Robotics, Computer Vision, Machine Learning, Industrial Automation, Embedded Systems.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB
Control & Estimation: LQR, MPC, SMC, PID, backstepping, H∞, adaptive & predictive control; state-space modelling; trajectory tracking (Pure Pursuit, Stanley, DWA)
Planning & Navigation: A*, RRT/RRT*, PRM, Hybrid A*, APF; SLAM (RTAB-Map); RVO multi-robot avoidance; PDDL / PlanSys2; ROS 2 Nav2
ML / RL: Reinforcement Learning (PPO), genetic-algorithm optimization, computer vision (OpenCV, MediaPipe); LLM / VLA integration
Robotics Frameworks: ROS / ROS 2, Gazebo, MuJoCo, CoppeliaSim, PX4 / MAVLink
Simulation & CAD: Fusion 360 / Inventor, Adams, Simscape, Simulink
Embedded & Hardware: Arduino, Teensy, Raspberry Pi, microcontroller programming, Embedded C, sensor integration
Operating Systems: Linux, Windows

PROFESSIONAL EXPERIENCE

Robotics Engineer, Humanoid Robotics Startup (Stealth) *Feb 2026 – Mar 2026*
- Built robotic-manipulation simulations in Gazebo and implemented PDDL task planning, letting task sequences be validated in simulation before committing to hardware.
- Simulated and tested a K-Scale humanoid in ROS 2 and stood up the sim-to-hardware deployment pipeline.

Research Assistant, Amrita Vishwa Vidyapeetham (under Dr. Rammohan Sriramdas) *Sep 2025 – Jan 2026*
- Developed a DST-funded two-wheeled hopping robot — characterized its jump dynamics and designed the stabilizing control, work published at ICRM 2025.
- Enhanced the control architecture and navigation performance of a quadruped mobile robot.
- Authored three research papers — one published, two under review.

Robotics Intern, Anvi Robotics *Jun 2024 – Jul 2024*
- Designed a water-cleaning unmanned surface vehicle (USV): component selection, CAD, and ROS simulations integrating MAVLink with PX4.
- Integrated vision-based trash detection and autonomous navigation in ROS, enabling the USV to locate and route to floating debris on its own.

Robotics Intern, Sony SSUP Program *Oct 2023 – Dec 2024*
- Co-developed a four-legged mobile robot for farm assistance and cattle health monitoring under the Sony SSUP program.
- Programmed and integrated multiple sensors and control algorithms, and troubleshot hardware to keep experiments running with minimal downtime.

Robotics Intern, 7s Technologies *Mar 2024 – Apr 2024*
- Designed an AR/VR motion-simulation chair with real-time control; selected components and ran load/performance calculations to hit motion-fidelity targets.

SELECTED PROJECTS

Vision-Based Drone Landing on a Moving Platform *Python, MuJoCo, CasADi*

- Built a **vision-only autonomous quadrotor** that lands on moving ground rovers and 6-DOF seakeeping ship decks from onboard sensors alone — ArUco fiducials fused in a relative-state EKF with a stabilized gimbal, no ground-truth state in the loop.
- Designed the control and safety stack: **geometric SE(3)** and nonlinear / tube-DOB MPC (CasADi), a **Hamilton–Jacobi reachability safety shield**, and higher-order control barrier functions for sense-and-avoid.
- Added reinforcement-learning (PPO / LSTM-PPO) landing policies and extended the system to a **decentralized multi-drone swarm** with Kalman-consensus estimation and a CBF-QP safety filter; backed by a 112-test suite.

OpenArm — Bimanual Manipulation Platform *Python, MuJoCo*

- Built a control, planning, and learning stack for the **bimanual Enactic OpenArm v2 (7-DOF × 2)** in MuJoCo: forward/inverse kinematics, Cartesian and **compliant (admittance) control**, and RRT-Connect obstacle avoidance.
- Implemented a range of manipulation skills — pick-and-place, articulated-object handling (drawers/valves), cloth folding, collision-aware bimanual hand-over, and dynamic ball-catching via **Kalman + MPC interception** — plus a learned ACT vision policy and an RL insertion suite.

NORMA — LLM-Supervised Robot Manipulation *SmolVLA, Claude, MCP, NormaCore*

- Ran a learned VLA policy (SmolVLA) and a reasoning LLM (Claude) **together on a real 8-DOF arm** — the LLM reasons over the camera scene and supervises the policy at runtime, taking over with corrected joint commands when it stalls or fails.
- Built LLM-driven pick-and-place (no hard-coded coordinates or IK), color sorting, and gesture-based Q&A, with safety enforced by a **custom MCP server** that gates every command so the model cannot execute an unsafe move. **Won the NormaCore track** at the HackLab Berlin AI × Robotics Hackathon.

Autonomous Quadruped Facility Inspection *ROS 2 Jazzy, Nav2, RTAB-Map, Gazebo Harmonic*

- Built a fully autonomous inspection mission for a **Unitree Go2 quadruped**: frontier exploration drives Nav2 to SLAM-map an unknown facility (RTAB-Map), the occupancy grid is segmented into rooms, and the robot visits each one with no operator input.
- Added open-vocabulary detection to find and 3D-localize equipment and auto-generate an inspection report; the same ROS 2 node graph runs unchanged in simulation and on the **real Go2 (sim-to-real)**. Built at the Europe Embodied 2026 Hackathon (Munich); backed by a 38-test CI suite.

sparsam — SLAM & Bundle-Adjustment Backend from Scratch *C++*

- Built a sparse nonlinear-least-squares / SLAM backend from first principles — Lie-group operations, analytic Jacobians, block-sparse Cholesky, fill-reducing ordering, and the **Schur complement** for bundle adjustment — with no dependency on existing solver libraries.
- Benchmarked directly against **GTSAM and Ceres** on standard public SLAM / bundle-adjustment datasets (BAL Ladybug, Carlone/Grisetti pose-graph suites), matching or beating their solution quality while running faster per iteration on the larger problems.

Swarm Autonomy — Decentralized Multi-Drone Search & Pursuit *ROS 2 Jazzy, PX4 SITL, Gazebo*

- Built a **decentralized multi-UAV** system for GPS-denied urban search-and-pursuit where each drone runs the same onboard stack — **visual-inertial odometry (OpenVINS)**, ESDF occupancy mapping (nvblox), planning, and control — with no central coordinator.
- Coordinated the swarm peer-to-peer over a modeled bandwidth-limited radio, allocating containment roles with **CBBA** to corner a fleeing target; runs full-stack on ROS 2 Jazzy + PX4 SITL + Gazebo Harmonic, backed by a 77-test suite.

Additional: AegisDrive — uncertainty-aware autonomous-driving safety platform; BallPark — classical-vs-learned mobile-autonomy benchmark; Comparative Control Study — 15+ benchmarked controllers, GA-tuned; Steady Stride — physical self-balancing robot with onboard manipulator; TurtleBot3 ROS 2 task planning (PlanSys2, PDDL, Nav2); manipdyn 6-DOF UR5e control & planning lab; quadrotor control benchmarks in Python & MATLAB.

PUBLICATIONS & RESEARCH

- M. R. Arumalla, R. Sriramdas, V. R. Devarakonda, A. Vincent, *"Mapping the Dynamics of Robotic Jumps in Hopping Robots,"* International Conference on Robotics and Mechatronics (ICRM 2025), 2025.
- Two additional manuscripts currently under review (2026).

ACHIEVEMENTS & LEADERSHIP

- **HackLab Berlin AI × Robotics Hackathon** (2026) — won the NormaCore track (team of 2) for an LLM-supervised robot-manipulation system on an 8-DOF arm.
- **Europe Embodied 2026 Hackathon** (Munich) — built an autonomous inspection system for a Unitree Go2 quadruped, sim-to-real deployed.
- **e-Yantra Robotics Competition 2022-23** (IIT Bombay) — awarded Level 3 for an autonomous pharma-delivery bot.
- **Robotics Club** — led the projects team; designed and delivered a hands-on ROS workshop.
- **NSS Volunteer** — contributed to community camps and social-outreach initiatives.